

Card Authorization Service Outage - Database Failover Delay

Meridian Commercial Bank - Internal Document

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Summary

On 2025-10-02, the primary database for the Card Authorization Service (CAS) experienced an unplanned failure. Automated failover to the replica took longer than expected, resulting in a full outage of card authorizations (both card-present and card-not-present) for 22 minutes.

Timeline

- 14:03 - Primary CAS database host becomes unreachable (underlying cloud provider host failure).
- 14:04 - CAS health checks begin failing; all authorization requests return HTTP 503.
- 14:05 - PagerDuty escalation to Payments on-call and Database on-call.
- 14:11 - Automated failover initiated but stalls due to replication lag (replica was 40 seconds behind at time of failure, exceeding the automated failover tool's 15-second safety threshold).
- 14:18 - Database on-call manually verifies replica data integrity and forces failover.
- 14:25 - CAS reconnects to new primary; authorizations begin succeeding.
- 14:26 - Full recovery confirmed via synthetic transaction monitoring.

Root Cause

Replication lag between the CAS primary and its standby replica exceeded the automated failover tool's safety threshold at the moment of failure, forcing a manual decision point. The lag itself was caused by a large batch reconciliation job that had been scheduled to run against the primary during business hours, generating an unusually high write volume.

Impact

- 100% of card authorizations failed for 22 minutes (14:03-14:25 UTC).
- Estimated 31,000 transactions declined-in-error.
- SLA breach: CAS availability SLA is 99.95%; this incident alone consumed the monthly error budget.

Resolution

- Manual failover executed by Database on-call.
- Batch reconciliation job rescheduled to a maintenance window (02:00 UTC).
- Replication lag alerting threshold lowered from 60s to 20s warning.

Action Items

1. Move all batch/reconciliation jobs affecting CAS primary to the nightly

maintenance window (Owner: Data Platform, Due: 2025-10-09) - COMPLETED.

2. Reduce automated failover safety threshold evaluation window, or add a force-failover-with-data-loss-acknowledgement fast path for on-call (Owner: Database Platform, Due: 2025-10-30).
3. Add synthetic authorization transaction probe at 10s interval (currently 60s) to reduce detection time (Owner: SRE, Due: 2025-10-16).

Related Incidents

See INC-2025-08-14-001 (Payment Gateway Timeout) - different root cause (connection pool exhaustion) but same downstream symptom class: card authorization failures.